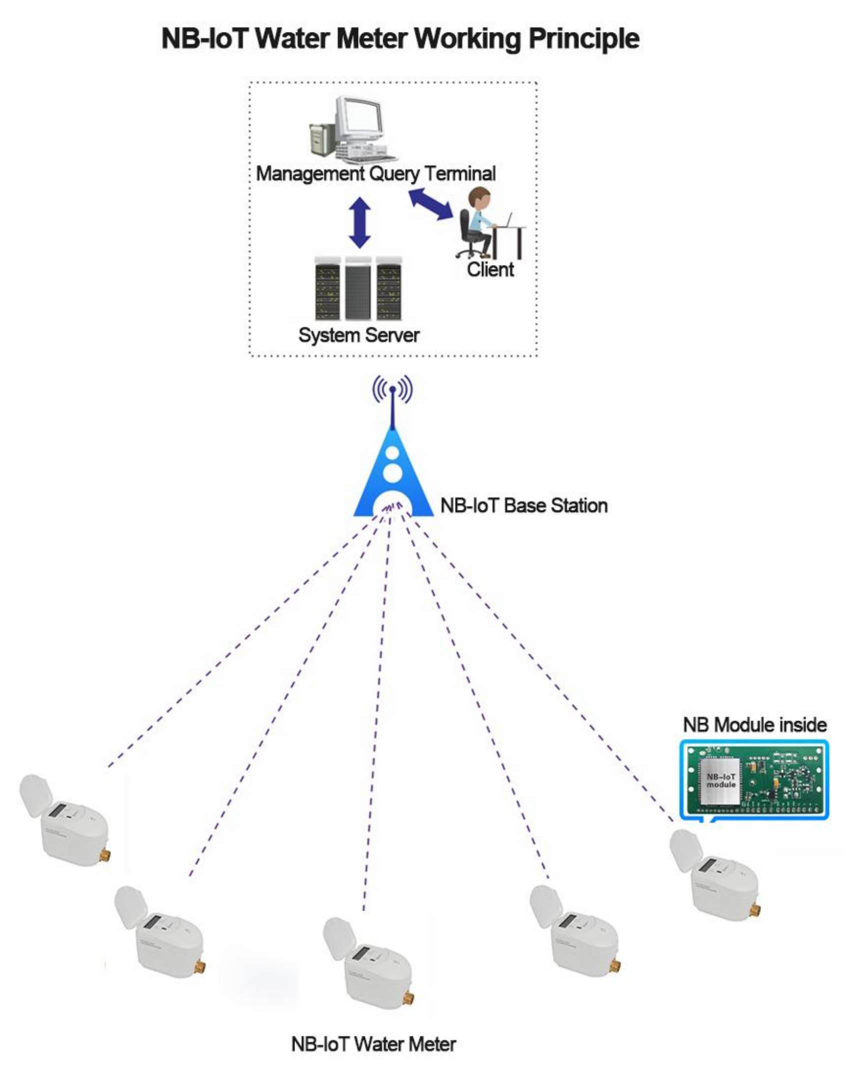
NB-IOT Ultrasonic Water Meter

User Manual

The NB-IoT ultrasonic water meter is one kind of smart water meter which is based on NB-IoT technology. We provide you the complete solution for the NB-IoT ultrasonic water meter. It can realize wireless remote meter reading and remote valve control.



Content

1. NB-IoT Ultrasonic Water Meter

1.1 Features

Ultra-low power consumption: utilizing capacity type battery ER34615H can be able to reach 8 years of lifespan;

Convenient to access: there is no need to rebuild the network, and it can be able to directly commercialized with the help of the telecom operator's existing network;

Large capacity: it can be able to storage 10 years of annual frozen data, 12 months of monthly frozen data, and 180 days of daily frozen data;

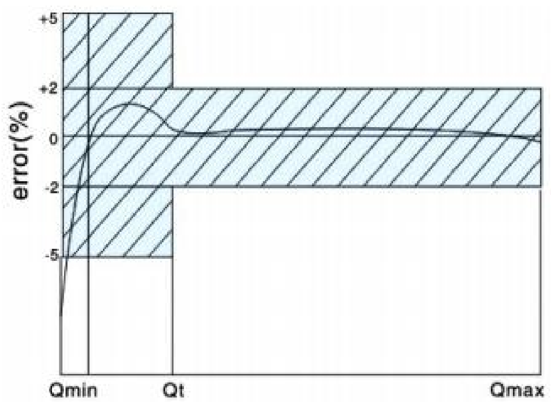
Technical parameters:

1. Standard: ISO4064
2. Body Material: Brass/Ductile iron/Plastic nylon etc4. Display: LCD screen
3. Connection Method: Thread/Flange
4. Display: LCD screen
5. R(Q3/Q1): R200/R250/R400/R500
6. Power Supply: Battery 3.6V
7. Lifespan of the Battery: 8 years above
8. Protection Class: IP68/IP65
9. Working Temperature: Cold water T30/T50; Hot water T90
10. Pressure Loss: △P≤0.63bar

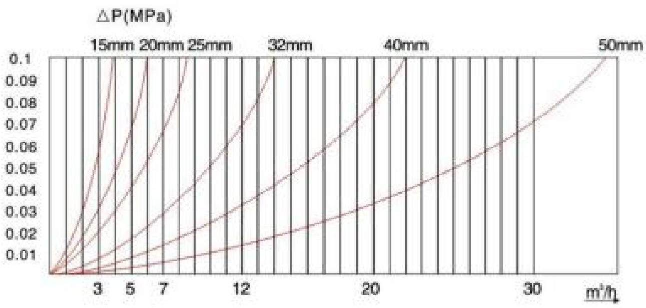
Metrology parameters:

DN	Ratio	Q1(m3/h)	Q2(m3/h)	Q3(m3/h)	Q4(m3/h)
15	R250	0.01	0.016	2.5	3.125
20	R250	0.016	0.0256	4	5
25	R250	0.0252	0.04032	6.3	7.875
32	R250	0.04	0.064	10	12.5
40	R250	0.064	0.1024	16	20
50	R250	0.16	0.256	40	48.75
65	R250	0.252	0.4032	63	76.781
80	R250	0.4	0.64	100	121.875
100	R250	0.64	1.024	160	195
125	R250	1	1.6	250	304.688
150	R250	1.6	2.56	400	487.5
200	R250	2.52	4.032	630	767.813
250	R250	4	6.4	1000	1218.75
300	R250	6.4	10.24	1600	19505

Flow error curve:



Data error curve:



A. Slow flow（Q1≤Q＜Q2）, Max permissible errors: ±5%

B. Water temperature≤30℃, Fast flow（Q2≤Q≤Q4）,

Max permissible errors: ±2;

Water temperature＞30℃, Fast flow（Q2≤Q≤Q4）,

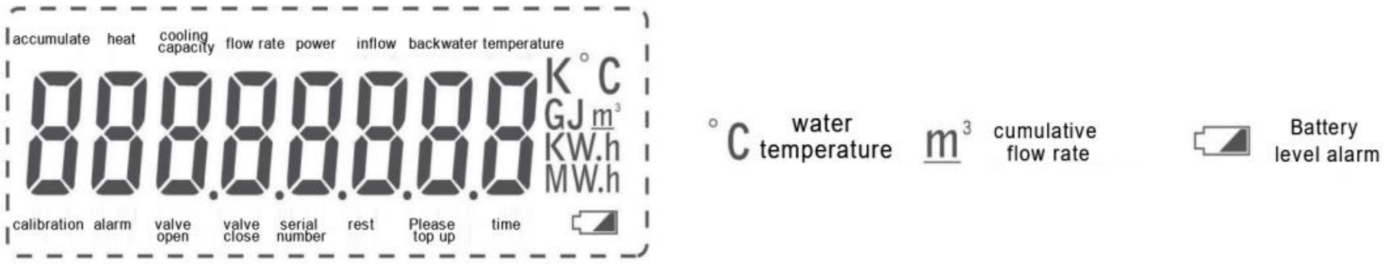
Max permissible errors: ±3%.

1.2 Using Instructions

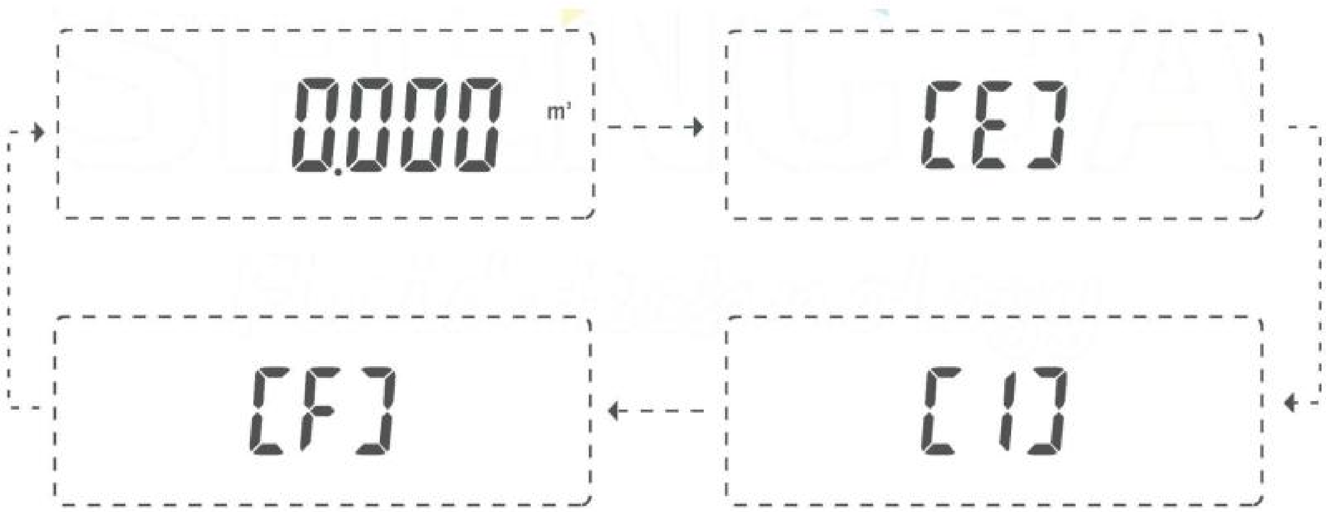
This water meter adopts an LCD screen with dual display of numbers and

English to present the measurement parameters and working status information to the

user, as shown below:



The housing to switch between different function menus. Operation and display consists of 4 groups of menus:



Arrow example

Indicates that the magnetic bar is used to touch the sensing area for more than 2 seconds and hold it (equivalent to a long press, hereinafter replaced by "long press")

Indicates that the magnetic bar touches the sensing area for about 1second and then moves it away (equivalent to a short press, hereinafter replaced by "short press"）

The 4 groups of menus are in the order of arrows:

1) Main Menu: Display menu during normal use

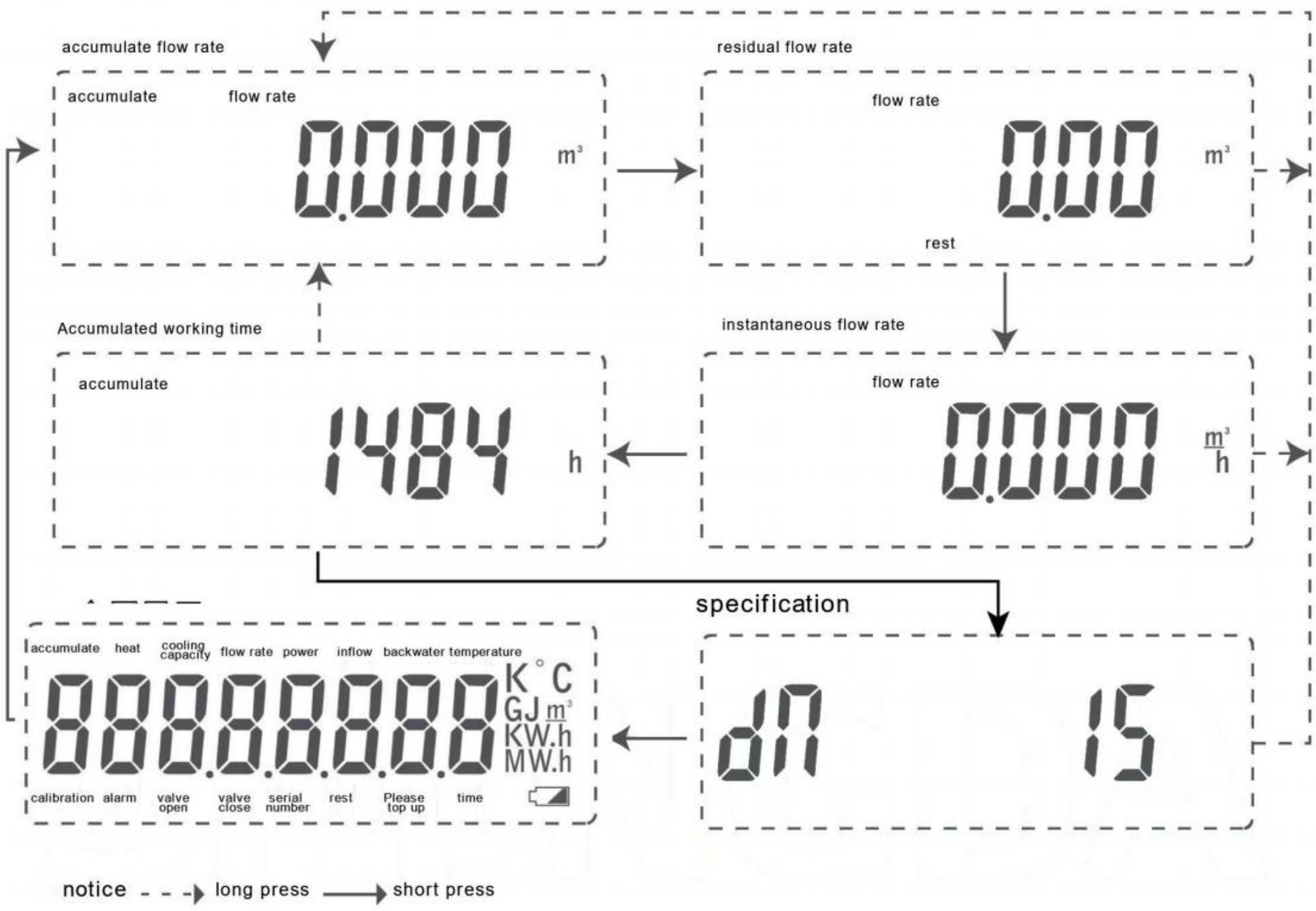
2) Fault display menu【E】:Displays the diagnosed fault with date and duration of occurrence

3) Information menu【I】:Display parameters such as water meter address and stored historical data

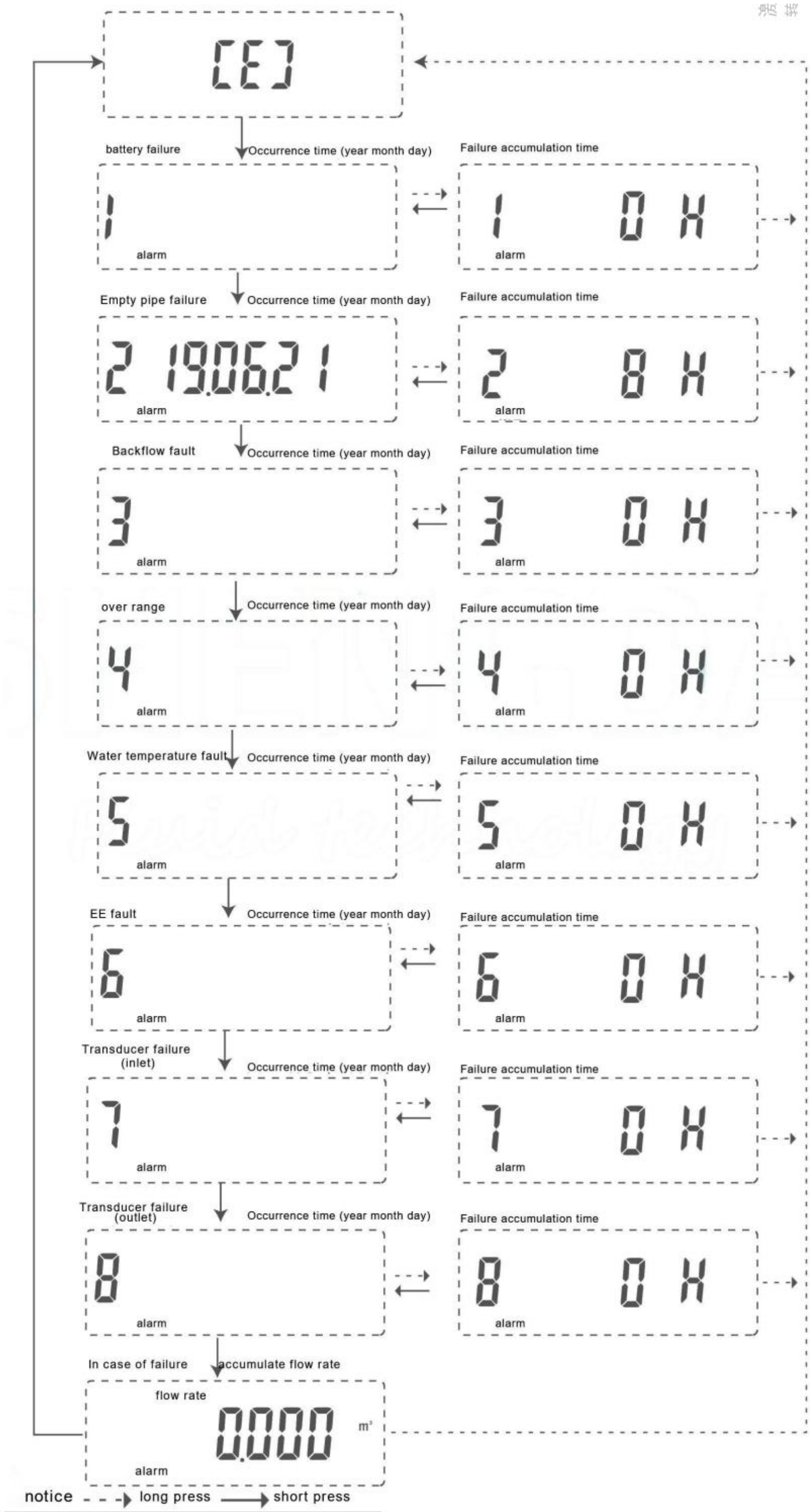
4) Detection menu【F】:Test only

The default setting of this water meter is that the LCD is always on.And the cumulative flow interface in the main menu is displayed fixedly.As shown in the figure above, the "long press" operation can switch in the above cycle; then the "short press" operation can be used to switch and locate the relevant display content in this group of menus.If you do not operate for more than 3 minutes, the LCD will automatically return to the main menu page (except when it is under the verification menu 【F】).

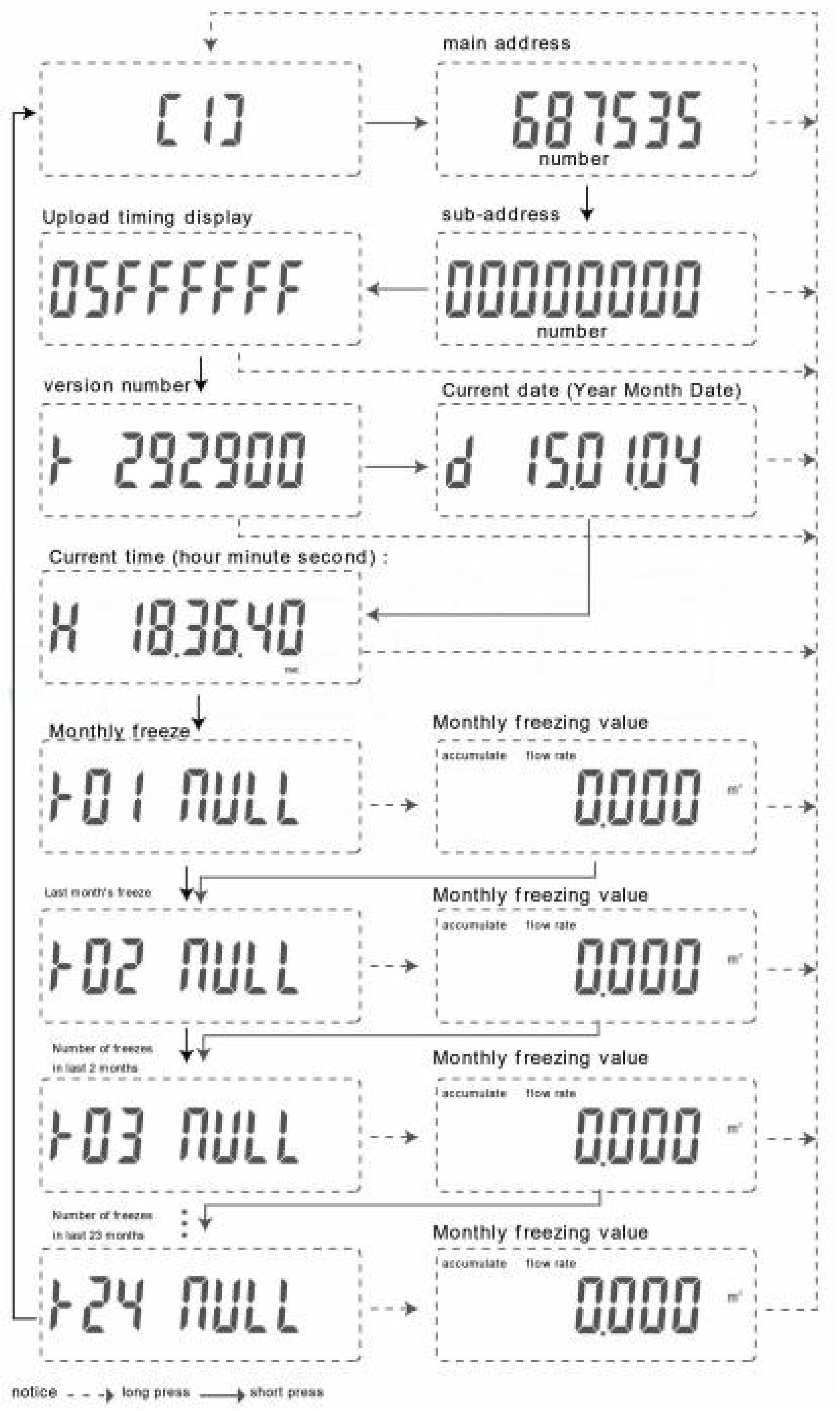
It should be noted that,Regardless of the menu display page (except the verification state), as long as there is water flowing through the water meter, the flow value will be automatically accumulated in the meter, and the measurement data will not be missed or undercounted due to viewing the menu content or key operation. The following is the cycle operation diagram of each menu：



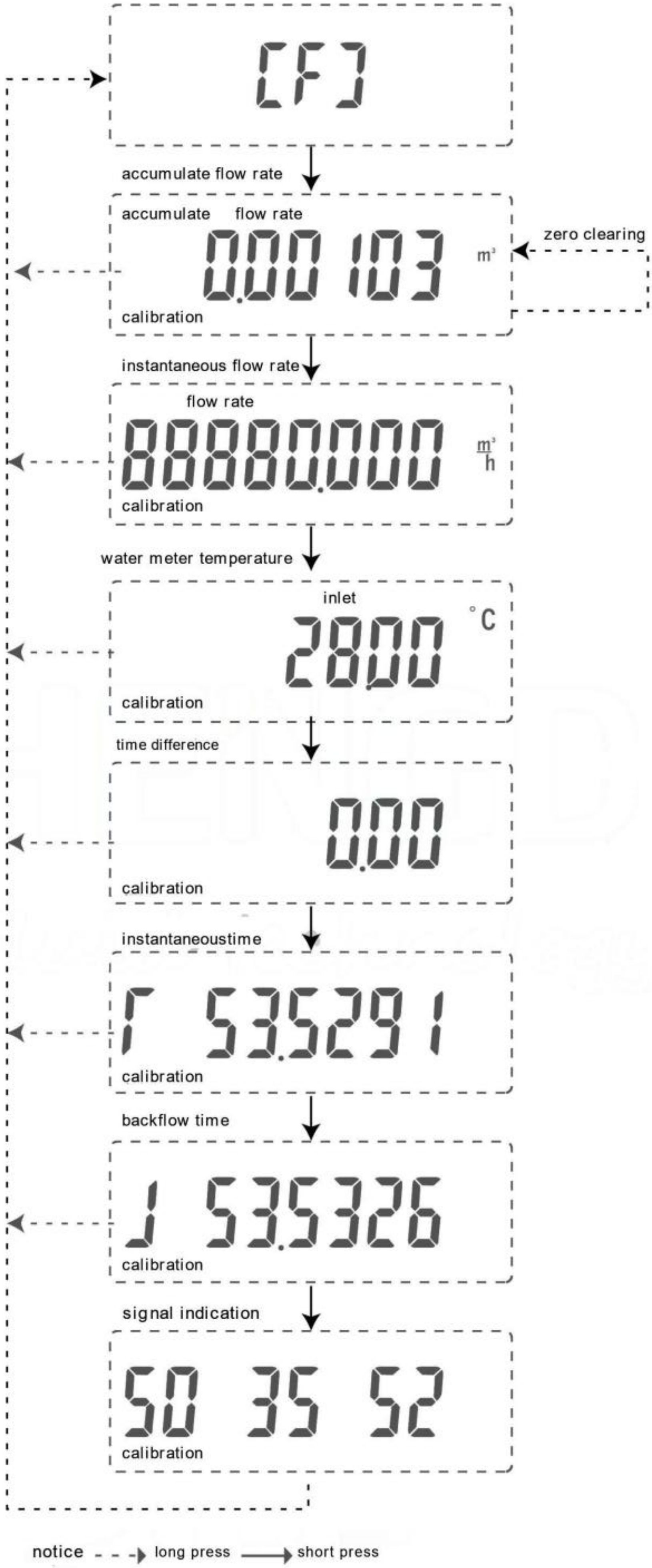
Fault Display Menu【E】



Information menu【I】



Detection menu【F】



1.3 Installation requirements for ultrasonic water meters

Special reminder: when the installed ultrasonic water meter is frozen in winter, please open at least one side of the valve (the valve in front of the water meter or the inlet valve and faucet in the user's room) in order to prevent the ultrasonic sensor damage caused by the pipe being frozen and cracked. This is particularly important if there is no occupancy or water.

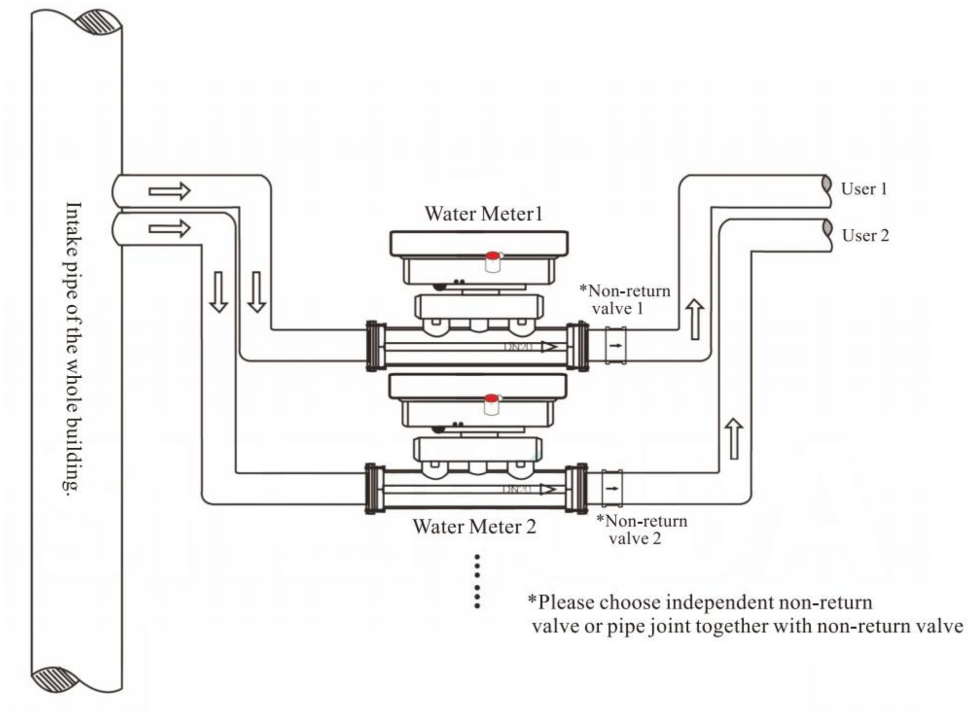
Because the measuring principle of ultrasonic water meter is different from that of mechanical water meter, the pipe should not be empty pipe or accumulate more bubbles, otherwise, ultrasonic signal cannot be transmitted, resulting in meter not counting or inaccurate measurement.

At the same time, in order to prevent the fluctuation of water retained in the measuring pipe segment due to the change of water pressure in the pipeline, which may affect the normal operation of the meter, it is strongly required to install check valve at the outlet end of the ultrasonic water meter. (Our company can provide pipe joint products integrated with outlet end check valve, which can be used together), Based on the above reasons, the recommended installation methods are as

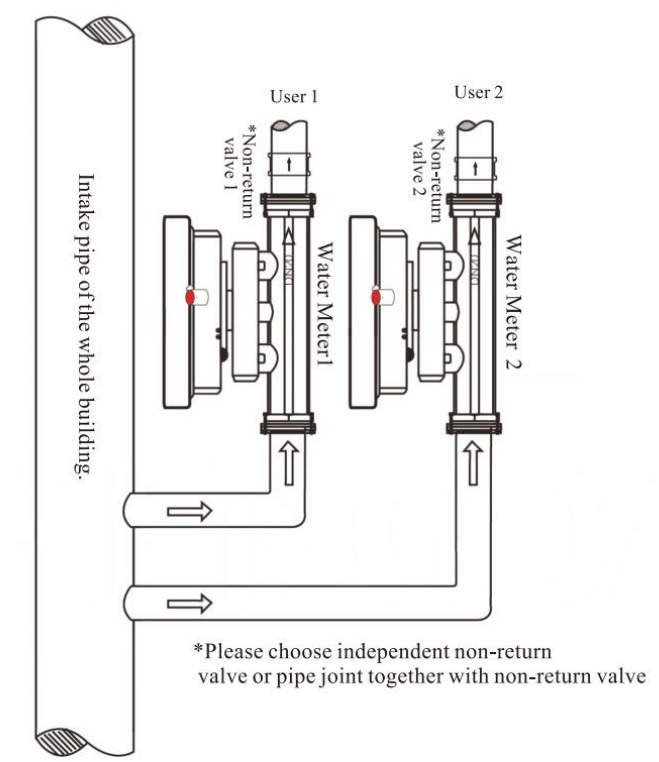
follows:

For horizontal installation, you are advised to install the pipes in the following shape:

U-shape. In this case, the ultrasonic water meter section at the lower part can remain full

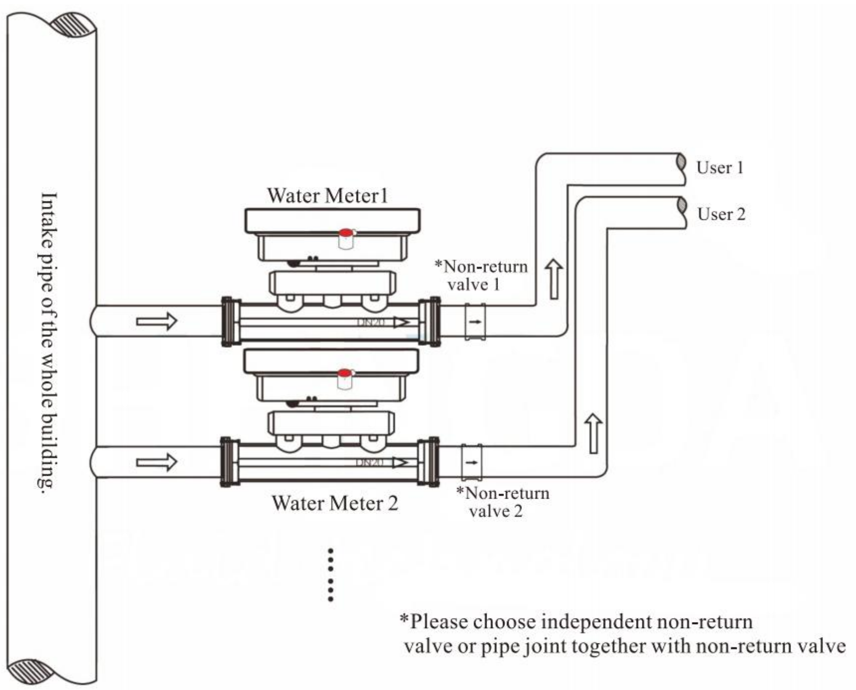


For vertical installation, as shown in the figure above, the water flow direction at the water meter is water inlet at the lower end and water outlet at the upper end. In this case, when there is water flow, it can avoid bubbles gathering in the water meter measuring pipe.



Compromise installation (horizontal)

When it is really difficult to implement the recommended horizontal installation conditions due to the objective conditions on site, the installation should at least follow the following figure:



In the figure, the pipe section in front of the water meter can be parallel to the body of the water meter (compared with the recommended method, the right-angle bend structure is canceled), but the pipe at the back of the water meter must be arranged as shown in the figure to avoid the accumulation of bubbles in the pipe.

PRECAUTIONS BEFORE INSTALLATION

1) Piping must be thoroughly cleaned before installing the ultrasonic meter to prevent debris from damaging the meter；

2) Ultrasonic water meter belongs to a more valuable precision instrument, pick up and put down must be careful, do not directly lift the table head or sensor line;
It is strictly forbidden to be near high temperature heat source (such as electric welding, to prevent battery explosion and injury and instrument damage)；

3) Special attention should be paid to the installation position of the ultrasonic water meter. It should be avoided to install the water meter at the upper end of the pipe (there will be bubbles in the pipe section), avoid to install the water meter near the elbow (will produce vortex flow), and keep it away from the pump and other equipment (will cause pulsating flow) ；

4) The connecting pipes of the upstream and downstream of the ultrasonic water meter should be consistent with the diameter of the water meter without reducing the
diameter；

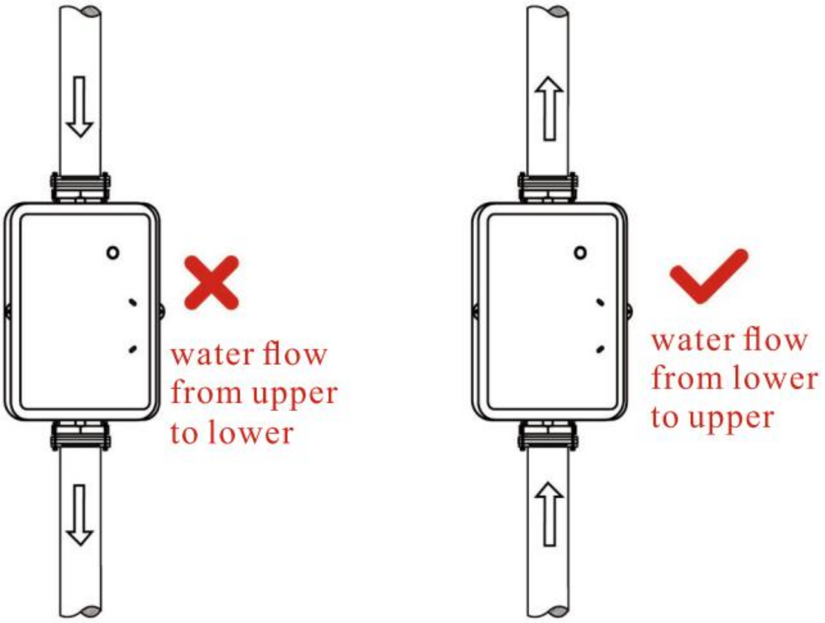
5) The direction indicated by the arrow on the surface of the ultrasonic water meter is the direction of water flow and shall not be reversed ；

6) It is recommended that the ultrasonic water meter be equipped with a filter of corresponding caliber before the meter. The front and back of the table are equipped with valves of corresponding caliber and can be separated from the table body for future maintenance and repair .

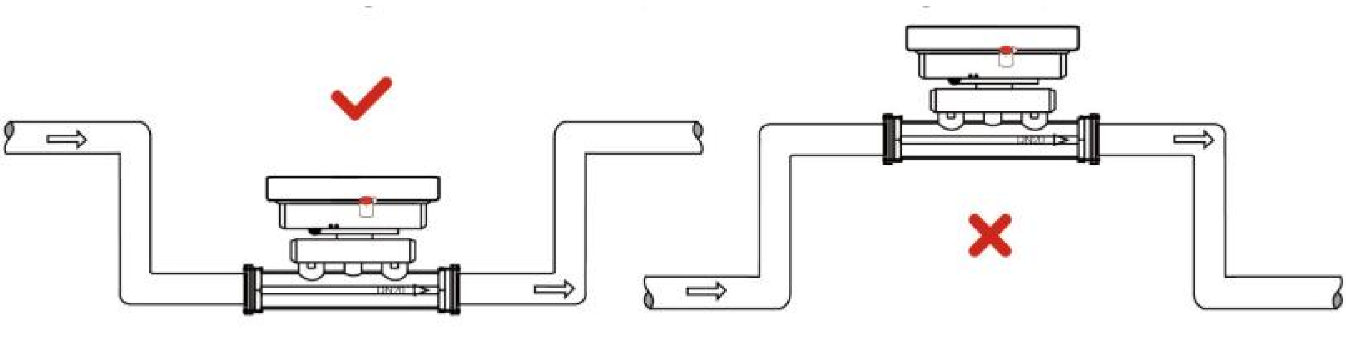
Common wrong installation examples

1) When the meter is installed vertically, it must be installed in the straight pipe with water flowing upwards, because the pipe with water flowing downwards will be unable to be filled with water under the action of

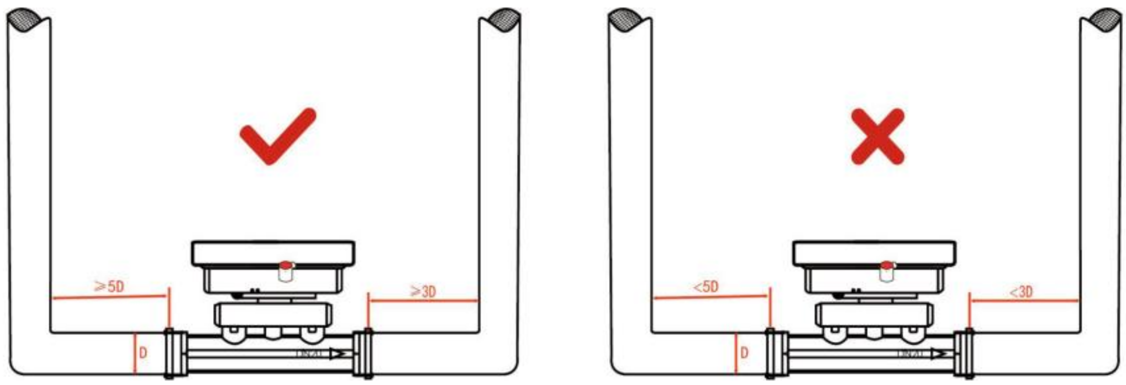
gravity, which will lead to inaccurate or even non-measurement of the meter (as shown in Figure C).



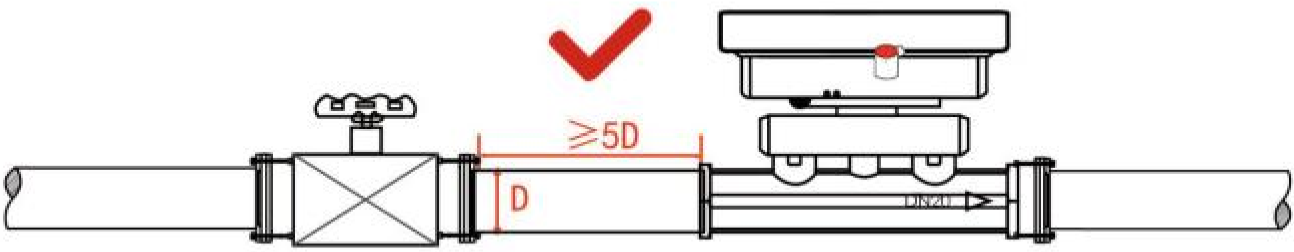
2)When installing the meter at the "U" shaped pipe, please install the meter at the lowest place, because air may accumulate at the high point of the pipe, resulting in inaccurate or non-metering of the meter (as shown in Figure D).



3) When the meter is installed at the elbow, the distance between the front straight pipe and the back straight pipe must be ≥5 times the pipe diameter, or the meter may be inaccurate (as shown in Figure E).



4) When installing valves or other objects in front of the table, it is necessary to ensure that the distance between the table and the object is ≥5 times the diameter, otherwise it may cause inaccurate measurement of the table; (As shown in Figure F)



2. NB Module of the Water Meter

2.1 Electrical Performance Parameters

Parameter	Min	Type	Max	Units
Working voltage	3.1	3.6	4.0	V
Working temperature range	-20	25	70	℃
Storage temperature	-30	-	80	℃
Sleep current	-	14.0	20.0	uA

Reminder: If the working temperature is continuously higher than 40℃ or lower than -20℃. The battery solution should be selected carefully. If you have any questions, please consult us.

2.2 NB-IoT Parameters

The working frequency band can be selected as telecom 850MHz (Band 5), mobile 900MHz (Band 8) or the NB-IoT frequency band supported by local operators, there is no need to apply for extra frequency points.

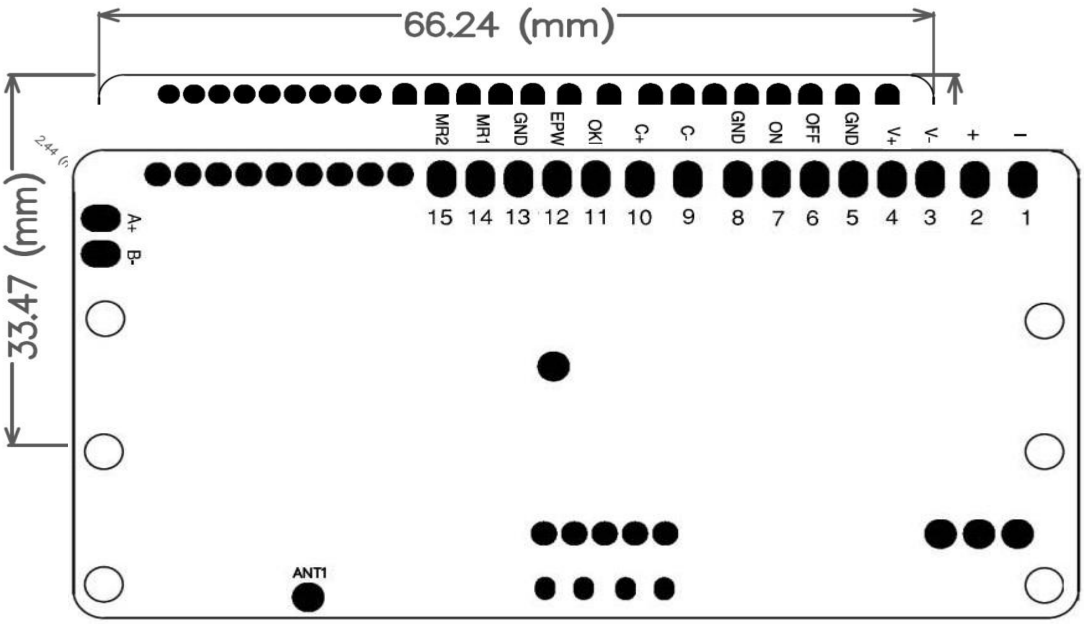
2.3 Communication Parameters

Infrared communication Rate：9600bps

Communication distance：0-15cm（Avoid direct sunlight）

2.4 Specification

Dimension



Pin Description

Item	Name	Description
1	-	Input power negative
2	+	Input power positive, 3.1V-4.0V
3	V-	Valve motor drive output terminal
4	V+	Valve motor drive output terminal
5	GND	Power negative, the signal can be able to refer the level output
6	OFF	Close valve in place detection
7	ON	Open valve in place detection
8	GND	Power negative, the signal can be able to refer the level output
9	C-	Composite capacitor negative terminal
10	C+	Composite capacitor positive terminal
11	OKI	Magnetic attack detection (the software can be configured to use)
12	EPW	3.0V controllable power output (the software can be configured for use)
13	GND	Power negative, the signal can be able to refer the level output
14	MR1	Metering test line 1
15	MR2	Metering test line 2

Description:

VDD: It connects to the positive of battery. It is recommended to use the capacity type battery ER26500+SPC1520.

EPW: The default is the DC power output terminal, which can be used to provide no more than 5mA working current for peripherals such as off-chip sensors. The current limit can be customized.

The remaining electrical interfaces on the module that are not described in detail in this document are temporarily not open to the public, please be sure to leave them in the air.

2.5 NB Module Function

2.5.1 Touch button

The button is one of the ways to make the module enter the maintenance mode. The function of the button is as follows:

Short press: don't exceed more than 2s, and the green led light will light up, indicating that the current terminal module is in an operational status.

Press key to trigger data report: press and hold until the red led light is on, and after letting go, touch the button again (with an interval of no more than 5s), and the green led light is on to activate the data report once.

The activation time lasts no more than 7min. If the operation fails, you need to wait 5s before the operation.

Enter maintenance mode: Press and hold until the red and green led lights are on at the same time, and then touch the button again after releasing your hand (the green led light is on) to enter the local configuration mode, and the window will be opened for 3 minutes. If there is normal data interaction, continue to open the window for 3 minutes, Exit after 3min without data interaction.

Silent time: The silent time is the duration of the module not responding to key actions. Each time during the data reporting process or during the infrared mode, the buttons are in silent state.

NB-IoT communication

The module interacts with the platform through the NB-IoT network. The main functions of NB-IoT module are as follows:

Regular data reporting: The terminal module regularly reports data according to the reporting cycle set by the user, once a day by default.

On-site trigger report: The report can be triggered by pressing the button or by the local command;

Multiple retransmissions:

According to the network conditions, the uplink data can be automatically retransmitted at 15~25min intervals for the data that fails to be reported regularly, and the maximum retransmission is twice, which effectively improves the reliability of communication;

NB-IoT communication protocol:

It usually adopts telecom COAP bottom communication protocol, it can also be switched to UDP protocol;

Support remote parameter configuration, parameter query, and time calibration;

Data reporting

Data reporting contains terminal information, base station signals, battery voltage, meter real-time data, dense data, frozen data, and alarm data etc.

Metering valve control

Support a variety of measurement methods, including single and dual pulse measurement (hall, reed switch, magneto-resistive sensor, etc.), non-magnetic, direct reading, etc., it's a fixed measurement method before leaving the factory.

Magnetic attack detection function, which generates an alarm sign when detecting malicious magnetic attacks.

Support the power off storage when the module is powered off, there is no need to re-initialize the measurement value.

Support remote valve control, the application platform can transmit commands to control the valve.

Support dredge valve function.

3. Management System

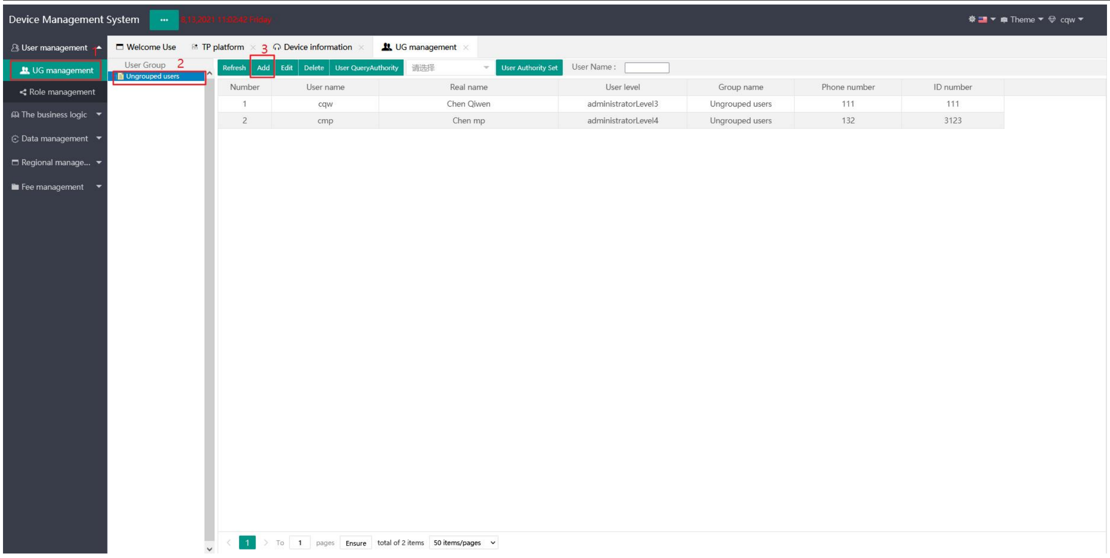
web login address: **http://www.haciot.cn:7007/mls/**

Step 1: Create sub-level users

If you do not need to create sub users, please skip to step 2 directly;

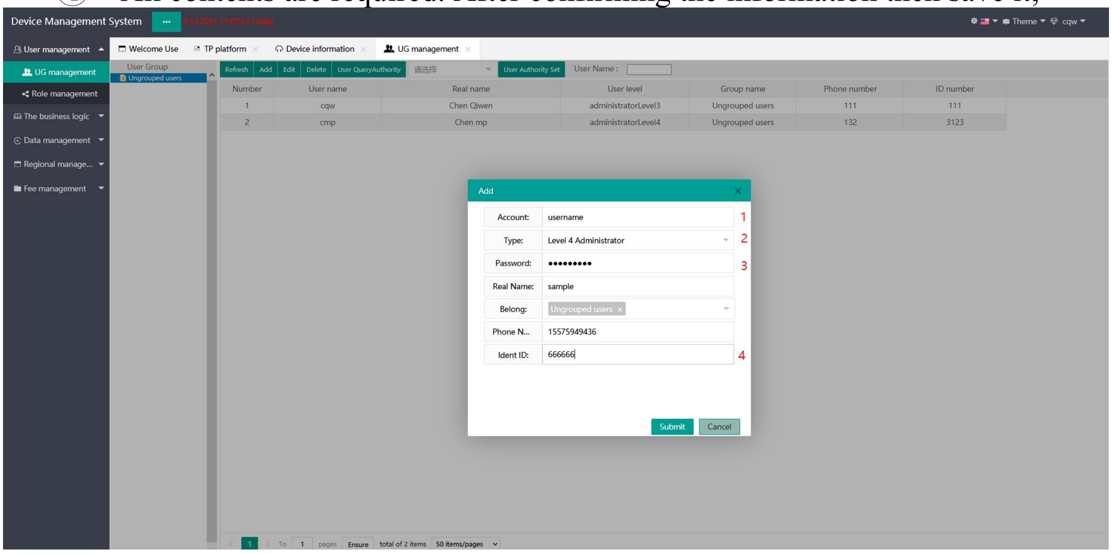
1. New users
* The current user can only create sub-level users (level 2 administrators can only create level 3 administrators and so on, up to level 5 ordinary users).
* All created users exist in the ungrouped groups, and are displayed according to the permissions of the currently logged-in user. By default, you can only view themselves and their sub-users.
* The user does not have permission and is granted permission based on the "role" function

① Click `UG management`
② Click `ungrouped users`
③ Click `Add` ,the create sub user dialog box will pop up



Fill in user information

1. User name only supports Chinese, English and numbers;
2. Level 2 administrators can only create level 3 administrators (level by level);
3. More than 6 passwords are recommended;
4. All contents are required. After confirming the information then save it;

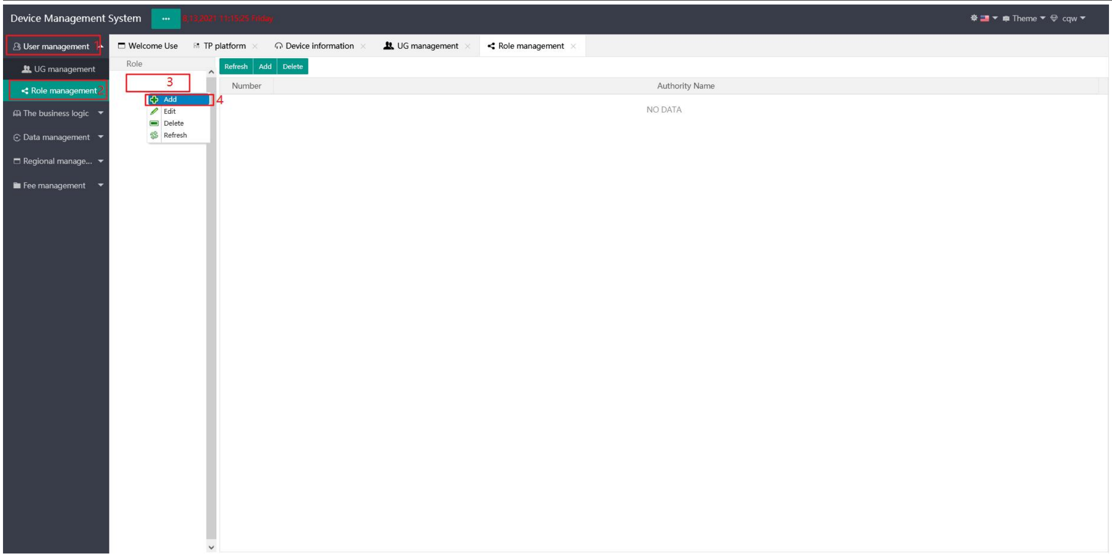


2. New role

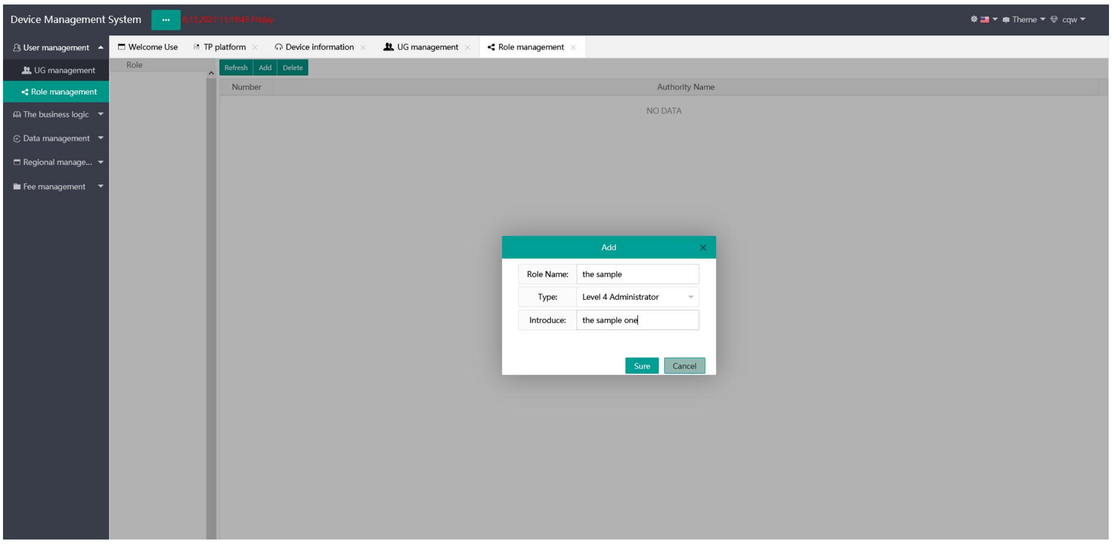
* There are operation permissions or roles to be assigned. Please skip this step directly

* Enter the role management page and right-click the new function on the role menu.

1. Click `User management`
2. Click `Role management`
3. Right click menu
4. Click `Add`

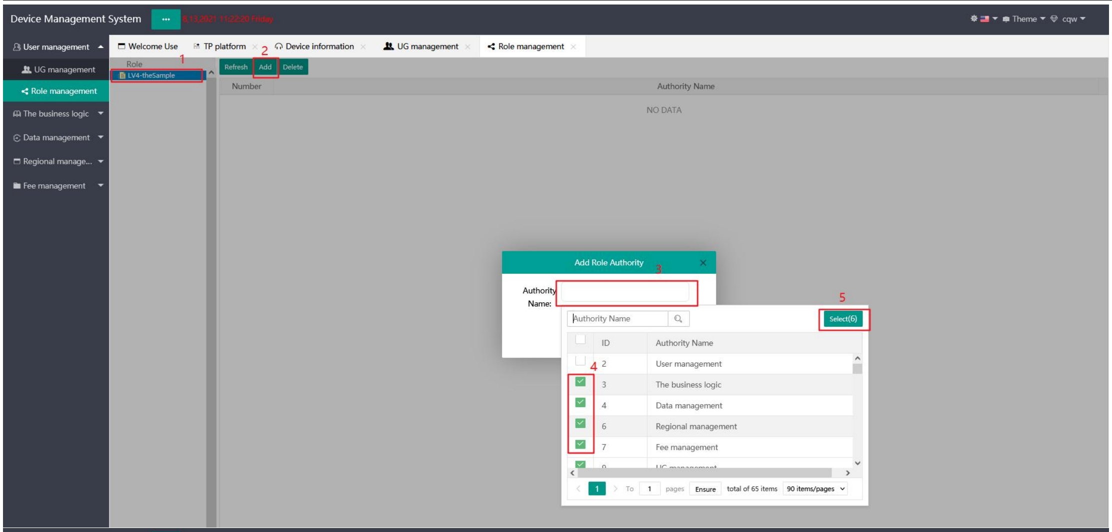


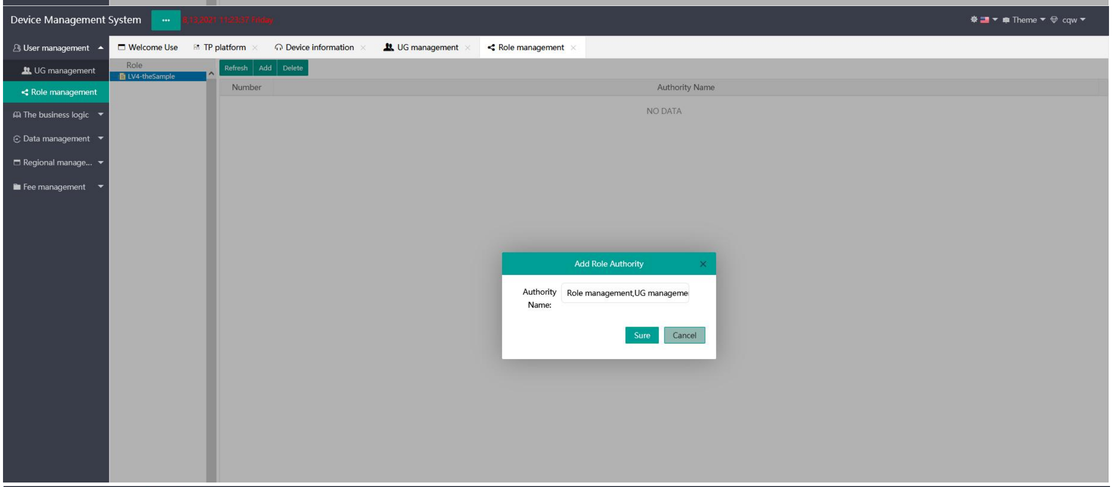
The role added here is used to manage the button functions of each page. Roles can be added according to customer needs.

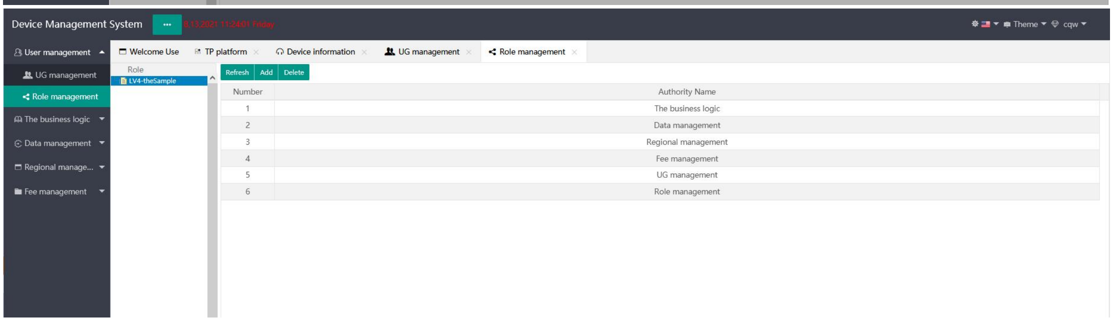


Click the Add button in the role permission list to add permissions to the role named "button".

① Click the newly created ` thesample`
② Click `Add`
③ Click `Authority Name` Input box
④ Check the required function permission
⑤ Confirm check





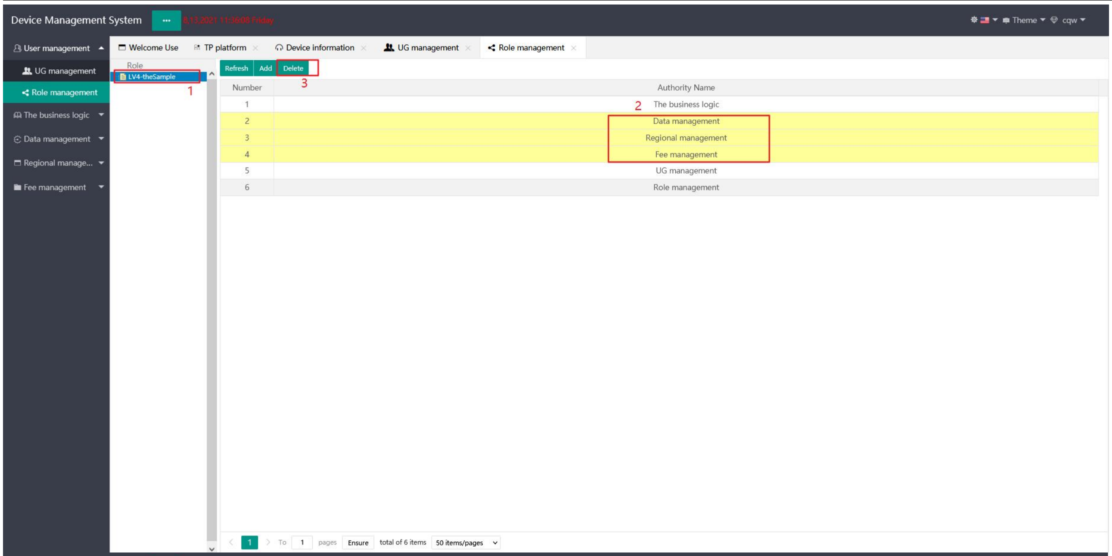


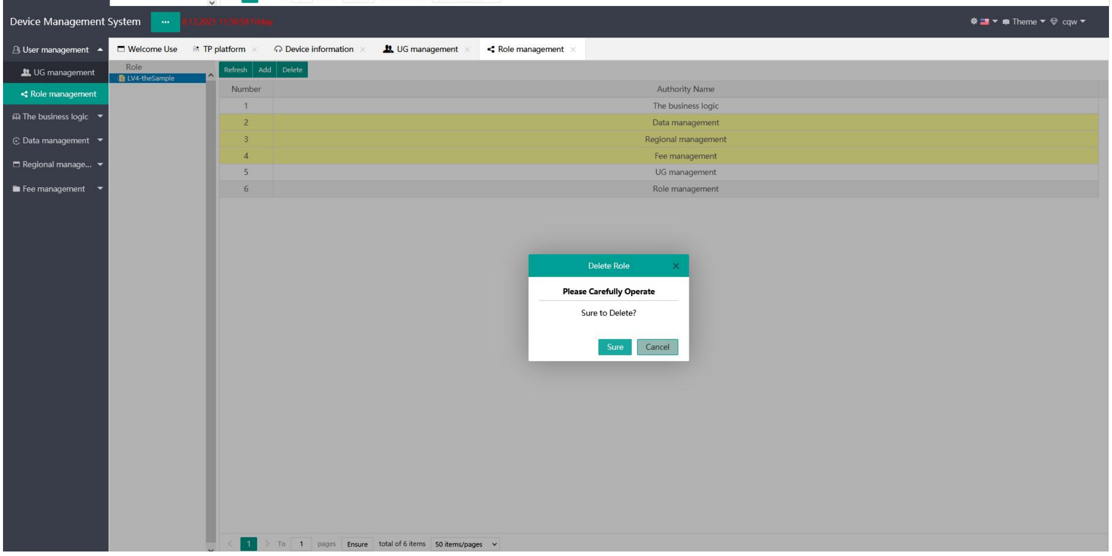
Role permission deletion

① Click `theSample`

② Click the function menu item to be deleted to support single click multiple selection

③ Click `Delete`

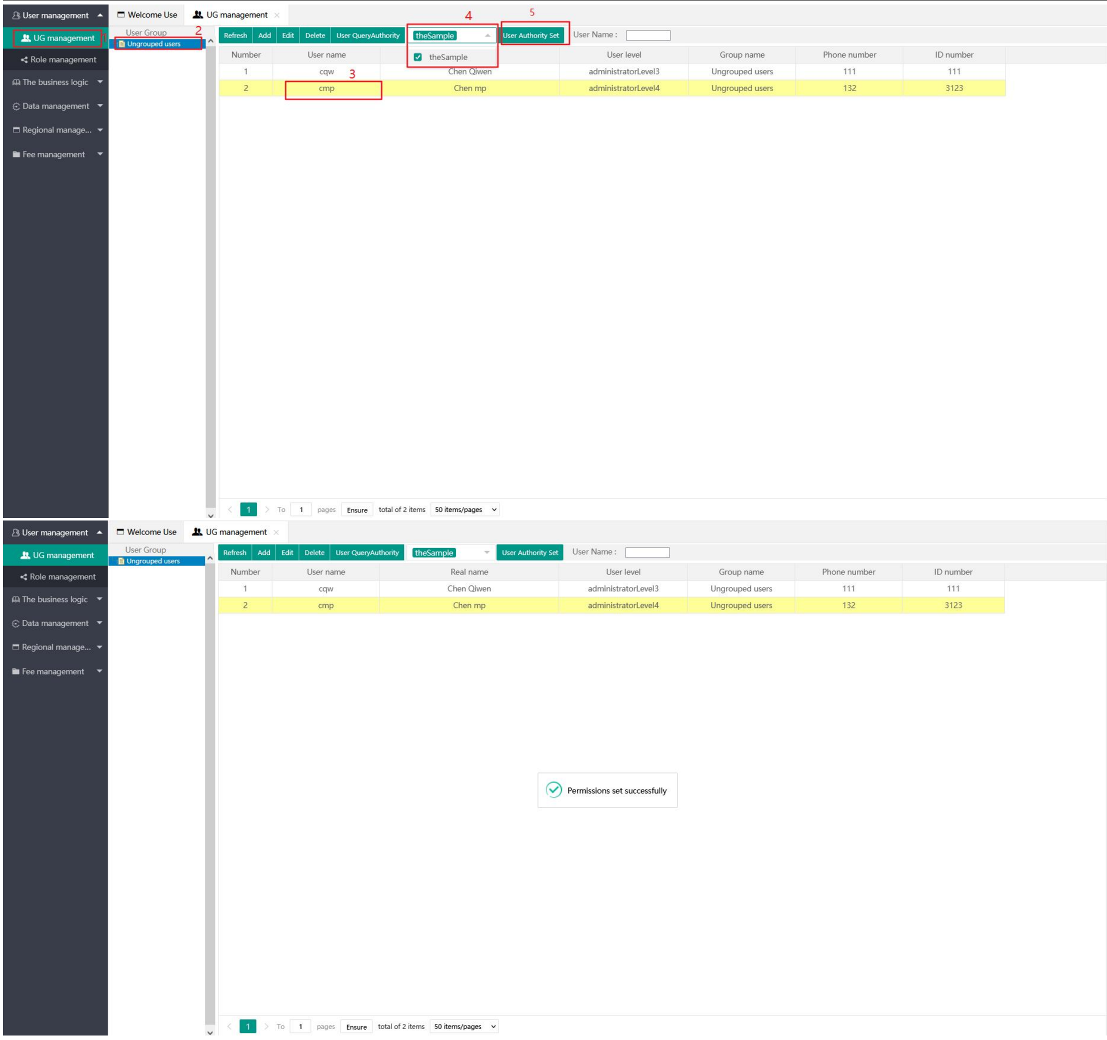




3. Role assignment (permission assignment)
* At present, there are two permission allocation methods: assigning role permissions; Assign group permissions
* Assign role permission: assign role permission can only be assigned individually and is limited by role level.
* Assign group permission: it is assigned to all member users in the user group and is not limited by role level. However, this group permission cannot be assigned to its sub users and can only be used by itself.

Assign role permissions

① Click `UG management`
② Click `Ungrouped users`
③ Click the user to be assigned `cmp`
④ Click the role selection box and check the role to be assigned
⑤ Click `User Authority Set`

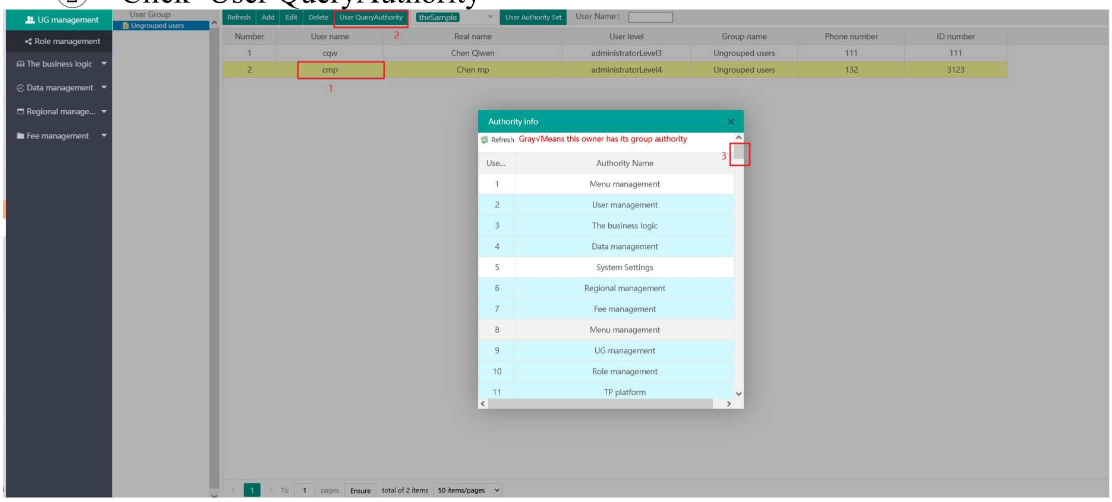


User authority query

* Query the menu and button permissions that users can access.
* Gray √ indicates group permission, dark √ indicates user permission, and unchecked means not owned.

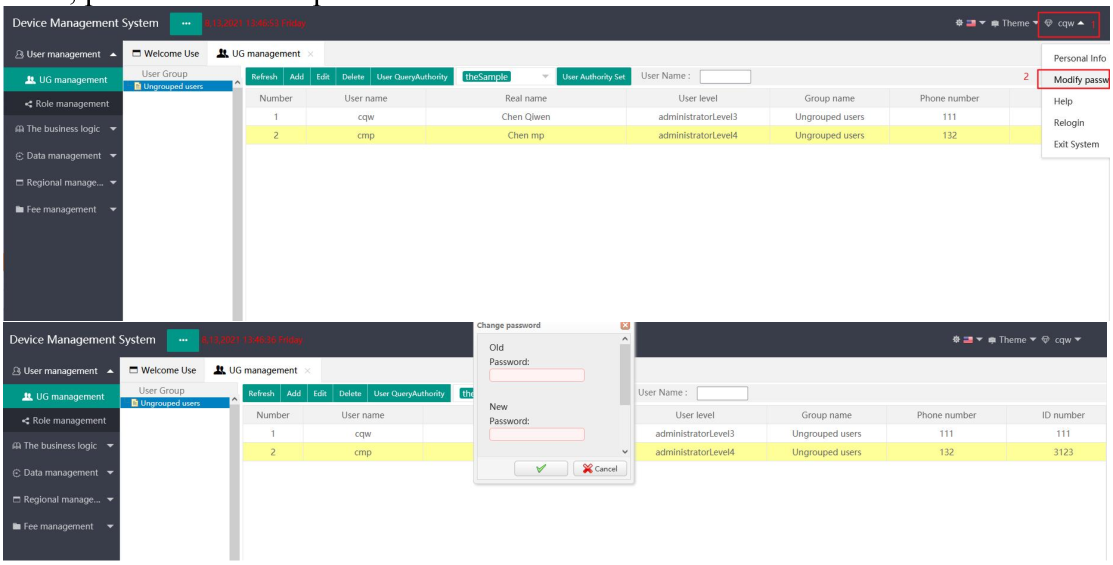
① Click user to be assigned ` CMP`

② Click `User QueryAuthority`



Modify user password

* Only the passwords of the current user and its sub users can be modified. To modify the password of the current user, please click personal information to modify the password, and to modify the password of sub-users, please use forced password modification

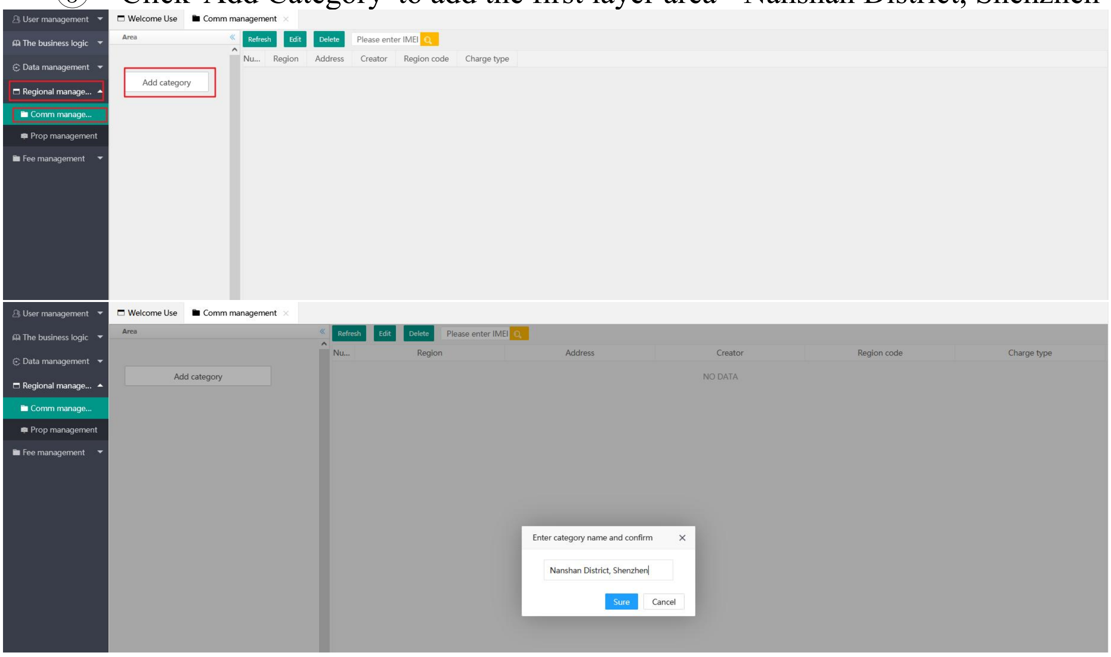


Step 2: Create a new area

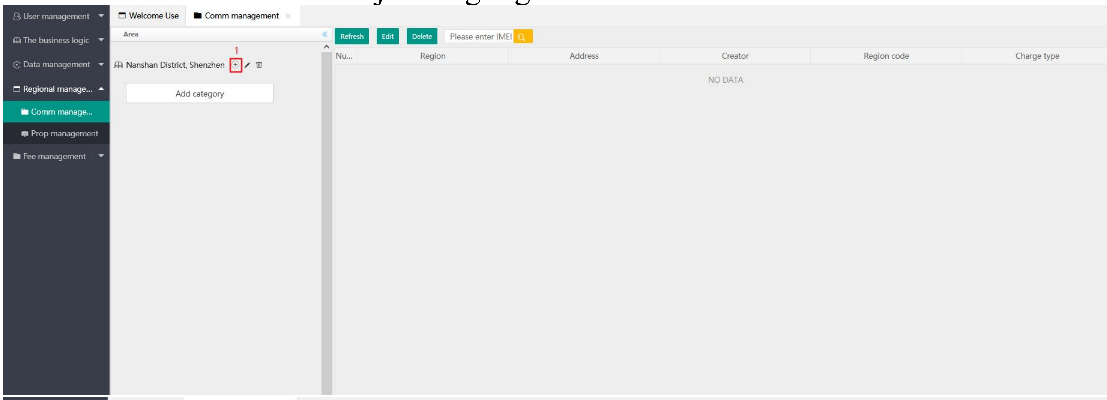
* Address information corresponding to the area, for example:Nanshan District, Shenzhen/ Xili Jiuxiangling Street/ Building B, the three-level area must be built, and all equipment information and owner information are in the third-level area.

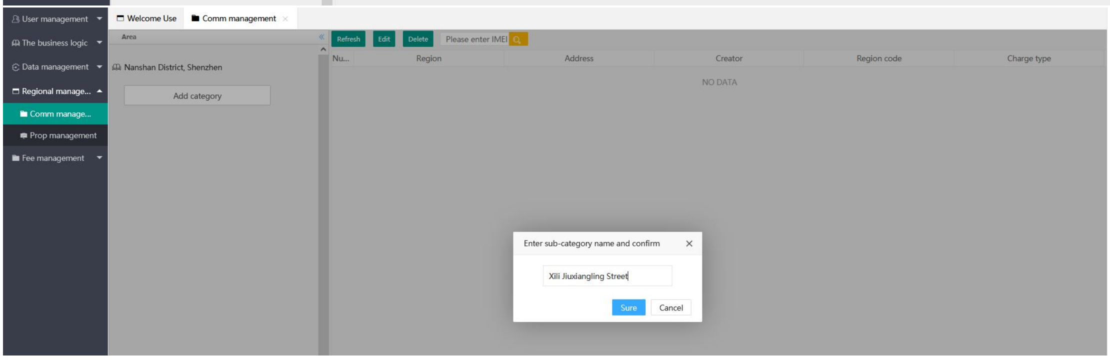
New first-level area

1. Click the 'regional management' main menu
2. Click the 'comm management' sub-menu
3. Click 'Add Category' to add the first layer area ` Nanshan District, Shenzhen`

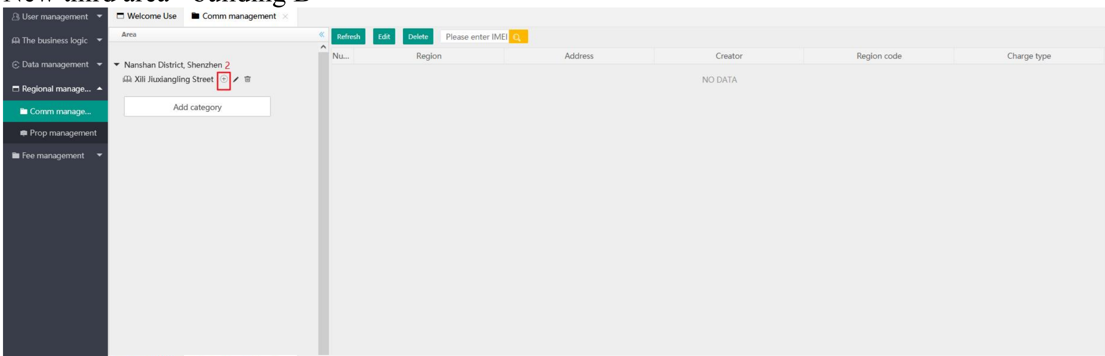


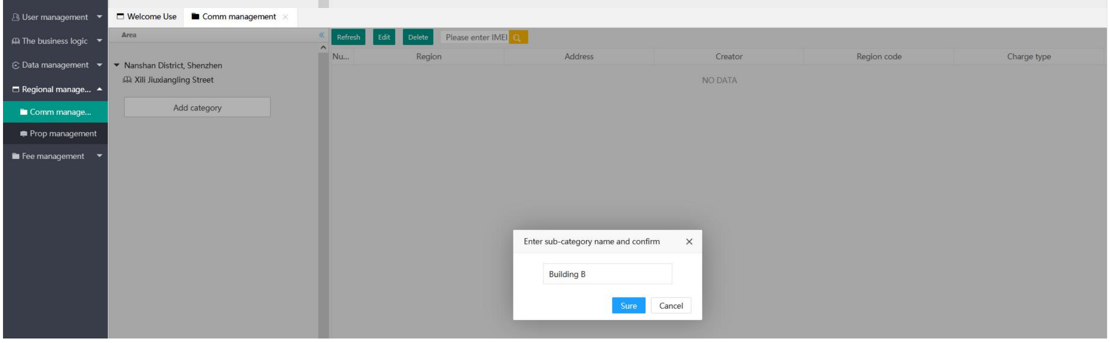
New second floor area ` Xili jiuxiangling Street`



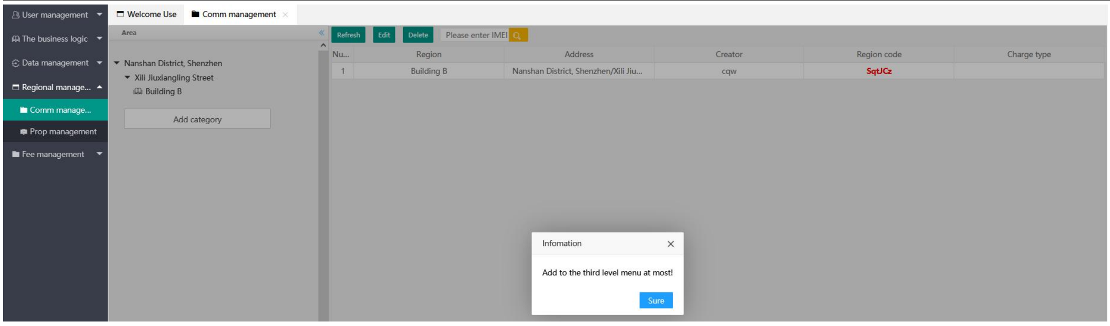


New third area ` building B`





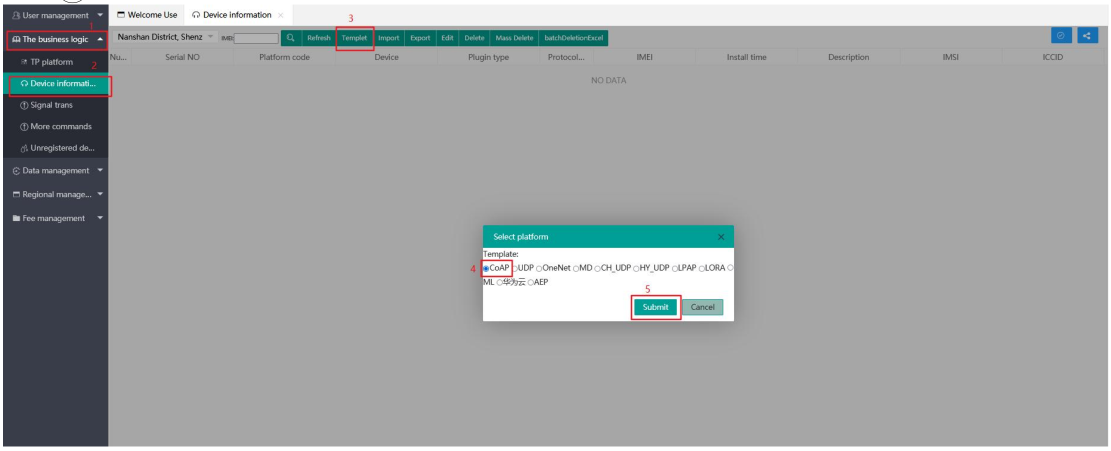
Only three regions are supported at most.



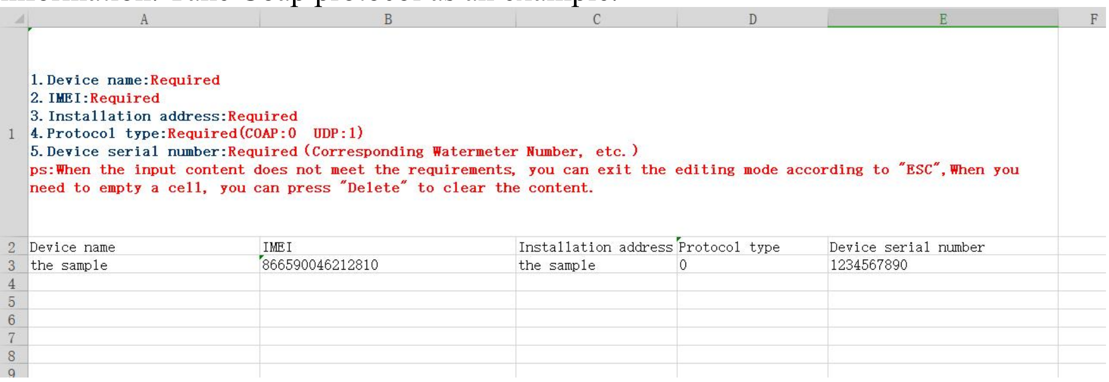
Step 3: Device registration

Enter the business logic menu device information sub-menu, click the template, and download the device corresponding to the agreement. After the download is complete, cancel the window.

① Click the `the business logic` main menu
② Click the 'device information' sub-menu
③ Click ` template`
④ Click the 'COAP' option
⑤ Click 'submit' to download



Fill in the template information, fill in according to the prompts, and pay attention to the correctness of the information. Take Coap protocol as an example.

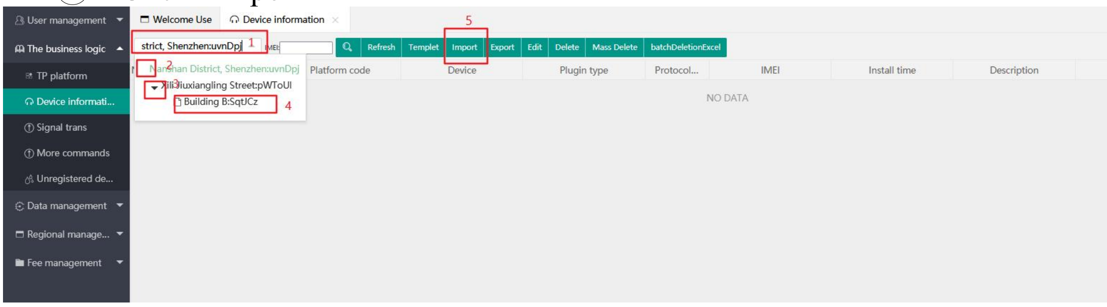


	A	B	C	D	E	F
1	1.Device name:Required 2.IMEI:Required 3.Installation address:Required 4.Protocol type:Required(COAP:0 UDP:1) 5.Device serial number:Required(Corresponding Watermeter Number, etc.) ps:When the input content does not meet the requirements, you can exit the editing mode according to "ESC",When you need to empty a cell, you can press "Delete" to clear the content.					
2	Device name	IMEI	Installation address	Protocol type	Device serial number	
3	the sample	866590046212810	the sample	0	1234567890	
4						
5						
6						
7						
8						
9						

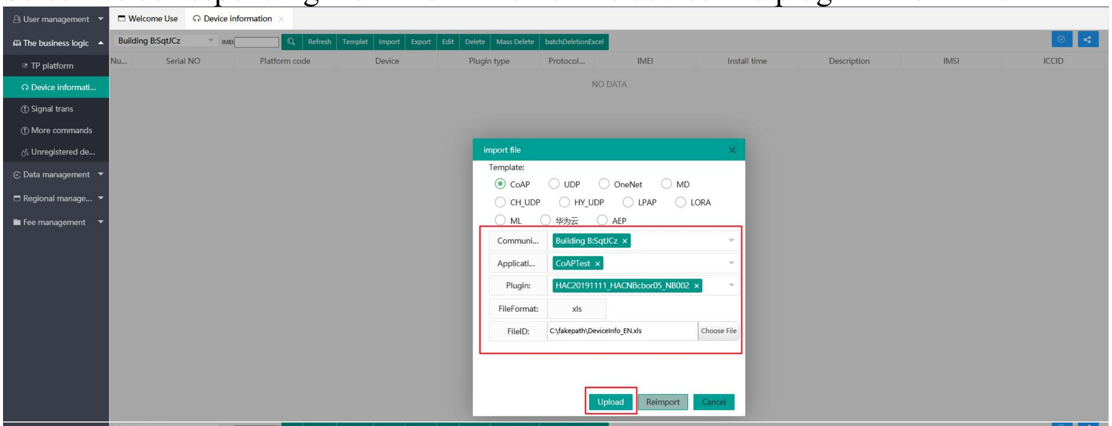
Open the import dialog

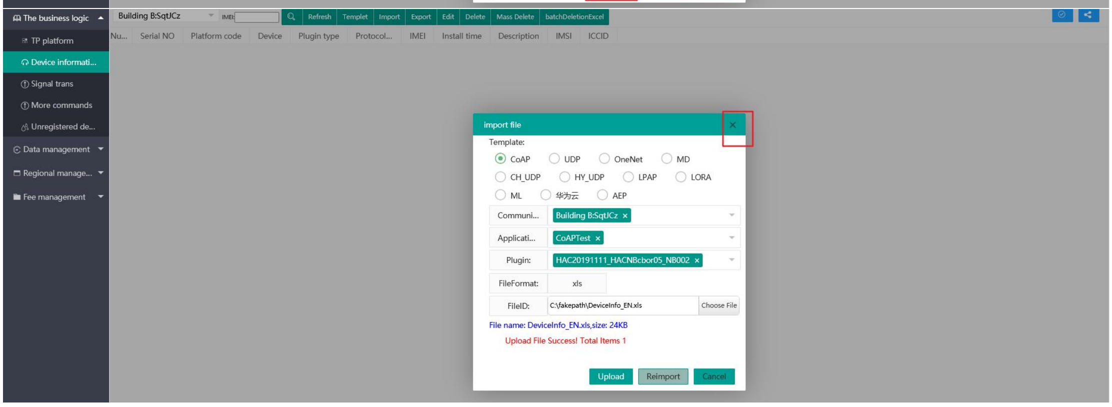
① Click the area selection box

② Click the front end to expand the second layer area
③ Click the inverted triangle to expand the third layer area
④ Click the third layer area
⑤ Click ` import`



Select the corresponding area information of the device and plug-in information

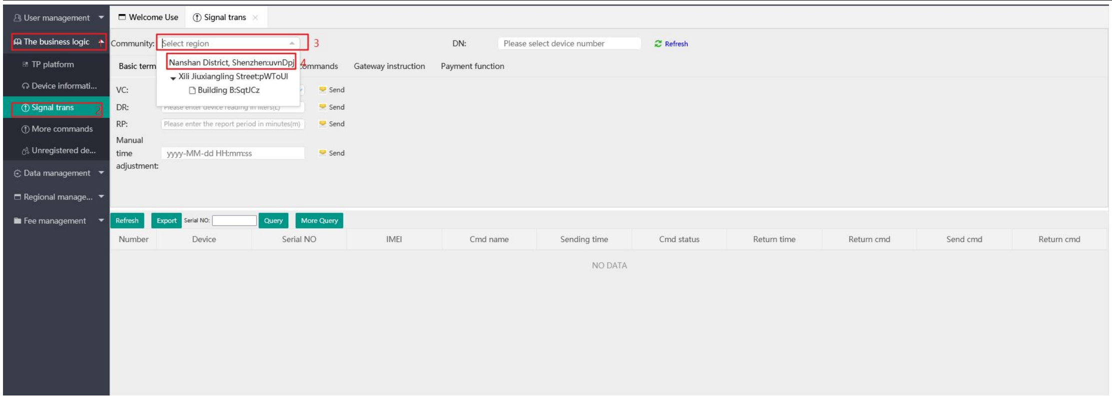




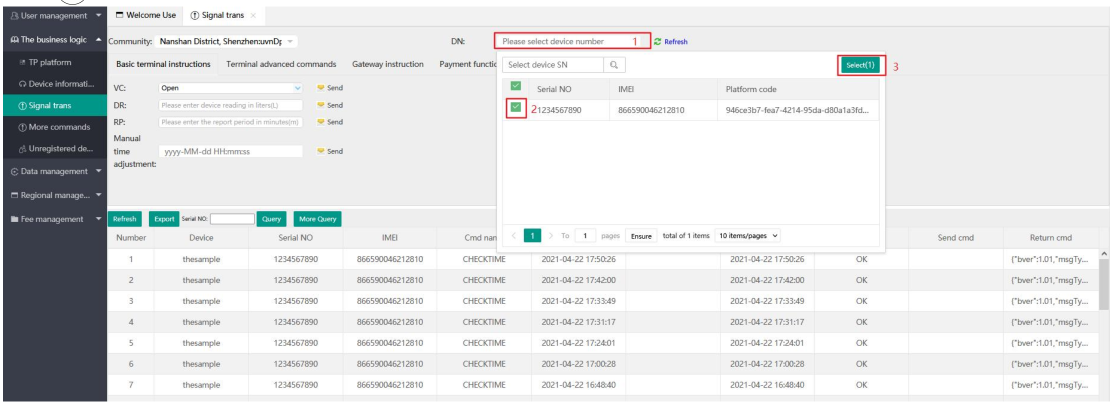
Step 4: Downlink control and data viewing

First select the area information, and then select the specified device information.

① Click the 'the business logic' main menu
② Click the 'signal trans' sub-menu
③ Click the 'community' selection box
④ Click the first layer area

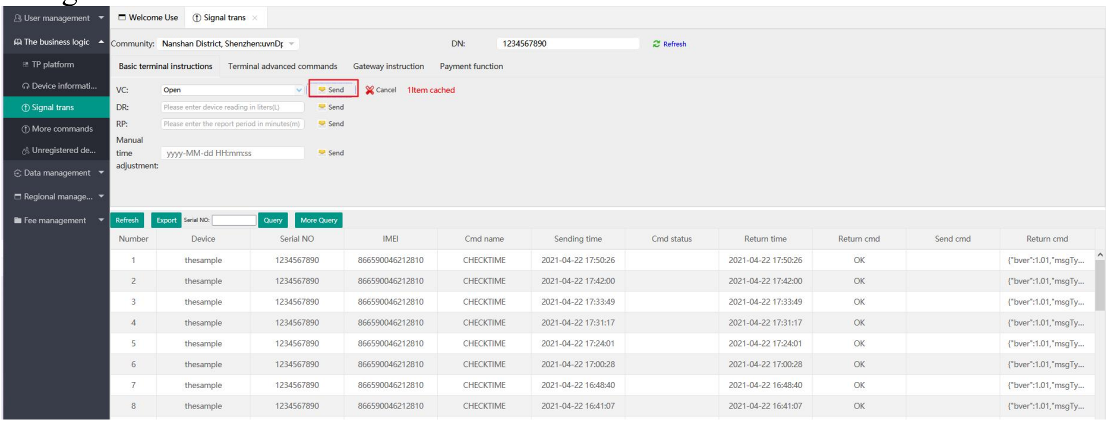


1. Click `DN(device number)`
2. Check the equipment to be controlled
3. Click 'select' to confirm

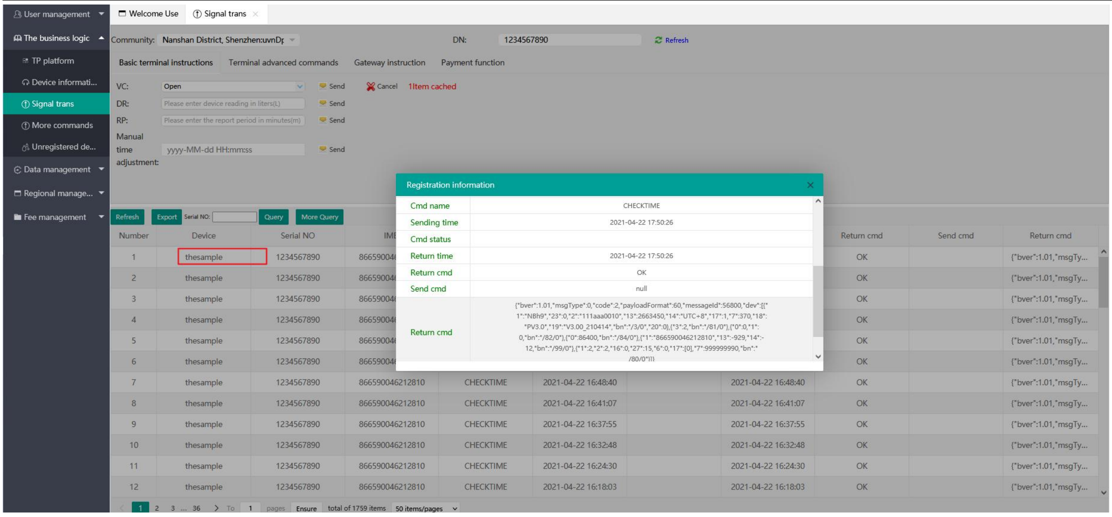


Choose to issue the instruction, click the send button, the instruction has been cached, and you need to manually trigger the device to receive it.

* The default cache time of the command is 48 hours, and then it is automatically cleared. The cached command must be issued after the previous command is issued and the response is pushed to the web background before the next command is issued.

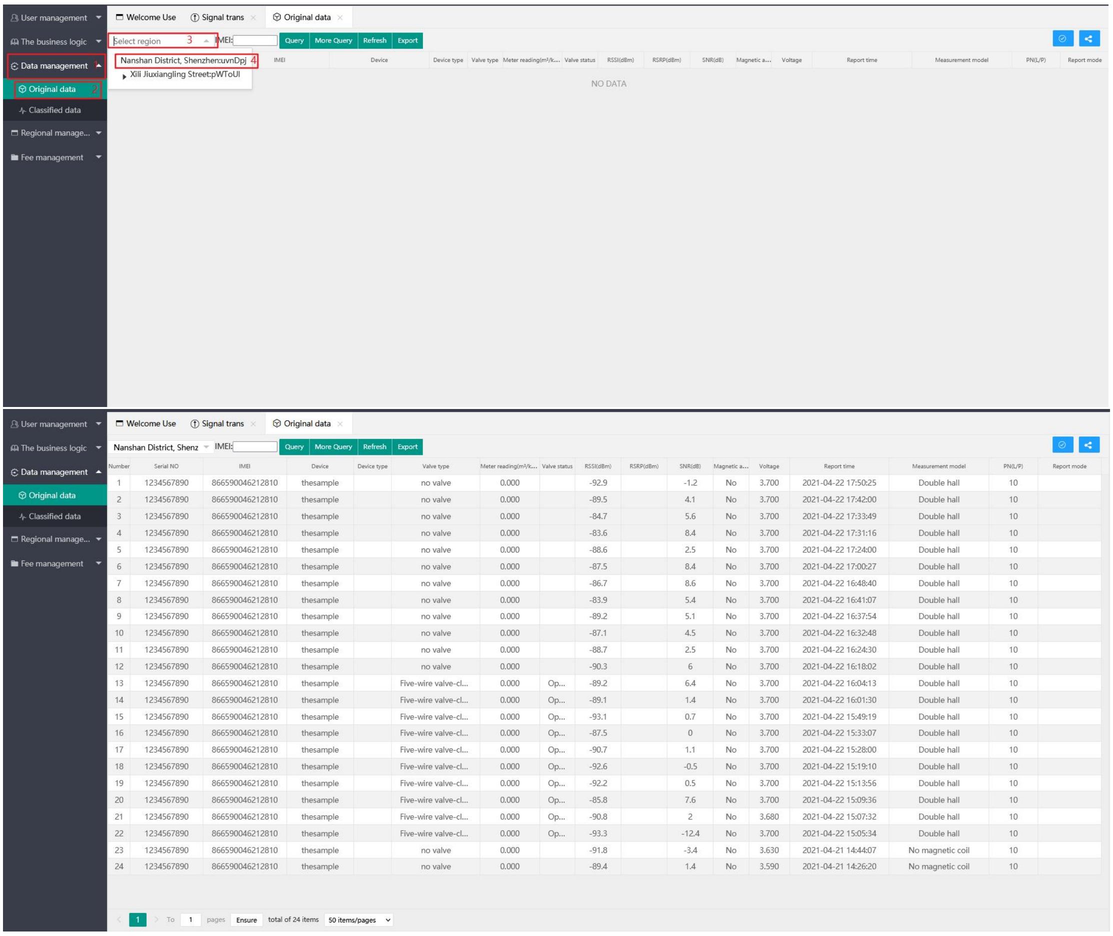


Double click the data information to query the details.

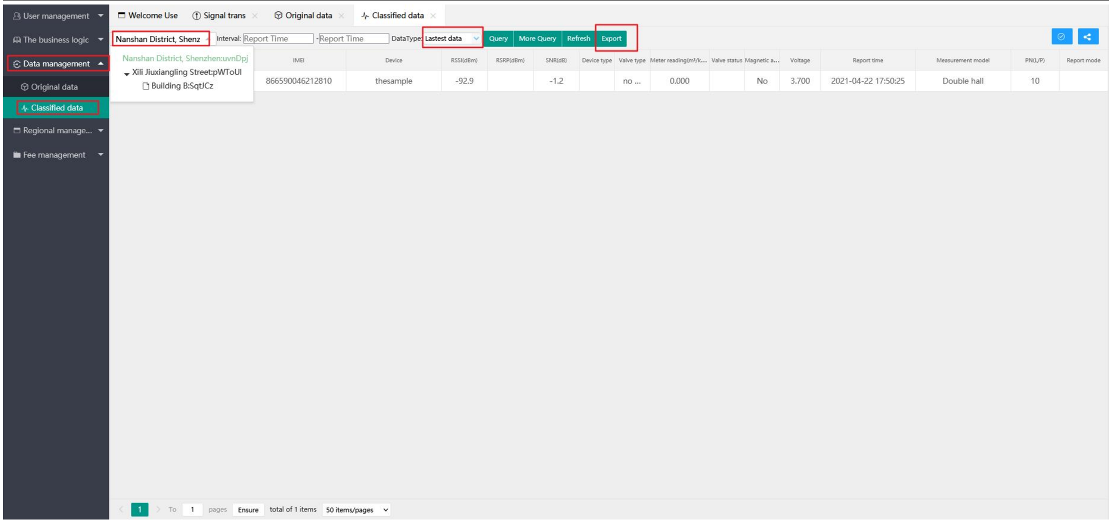


View the data reported by the device.

1. Click the 'data management' main menu
2. Click the 'original data' sub-menu
3. Click the 'community' selection box
4. Click the first layer area



Classification data export


User management
The business logic
Data management
Original data
Classified data
Regional manage...
Fee management
Welcome Use
Signal trans
Original data
Classified data
Nanshan District, Shenz
Interval: Report Time -Report Time
DataType: Lastest data
Query
More Query
Refresh
Export
Nanshan District, Shenzhen:uvnDpj
Xili Jiuxiangling Street:pWToUI
Building B:SqtJCz
IMEI
Device
RSSI(dBm)
RSRP(dBm)
SNR(dB)
Device type
Valve type
866590046212810
thesample
-92.9
-1.2
no ...
0.000
No
3.700
2021-04-22 17:50:25
Double hall
10